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Does Propeller Performance Depend on Airfoil Design?

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Airfoil geometry is an important feature for wing design. Wrong airfoil might cause a degraded performance to a winged air-vehicle. Propellers are operated as “rotating wing”, thus one might consider the same importance of airfoil upon propeller performance. The paper tries to inspect this misleading paradigm by presenting the impact of propeller air-section’s 2-D aerodynamic characteristics, over its performance. Results show that the impact of airfoil drag is very small. Even on cases, where the basic sections are sub-optimal, such as low Reynolds regime, radical 2-D aerodynamic improvement causes low performance impact. It is obvious that lift-to-drag ratio per-se is not the only criteria for airfoil selection. Still, it is the most prominent over propeller performance. Current effort proves the relatively low importance of this feature. Other airfoil considerations are mentioned and in many cases are much crucial for propeller’s airfoil selection.

I. Nomenclature

A	=	disk area
AF	=	activity factor
c	=	chord
c_{av}	=	average chord
C_d	=	2-D drag coefficient
C_{d0}	=	equivalent 2-D drag coefficient
C_l	=	2-D lift coefficient
C_{l0}	=	design 2-D lift coefficient
C_P	=	power coefficient
C_T	=	thrust coefficient
FM	=	figure-of-merit
J	=	advance ratio
M	=	Mach number
N_b	=	number-of-blades
P	=	total required power
P_{Cd}	=	cross-sectional drag power
r	=	radial coordinate
R	=	radius
Re	=	Reynolds number
t/c	=	thickness ratio
T	=	thrust
v_a	=	induced velocity
$v_{a,h}$	=	hover induced velocity
$v_{a,ideal}$	=	ideal induced velocity
V	=	airspeed
V_{tip}	=	blade tip-speed at static condition
ρ	=	air density
η	=	efficiency
κ_a	=	induced-velocity factor
σ	=	solidity
Ω	=	rotational speed

II. Introduction

Propellers are rotating wing in nature. This lesson was learned early in the flight history and implemented by the Wright brothers.¹ As such, every wing cross section seems to influence its characteristics. Hence, designing a wing, starts with careful choice of the air-section. This is not the case with propellers, as will be shown in the current effort. Literature is packed with example to this claim.

For example Moffitt et-al² discussed thoroughly the importance of inaccuracies of various propeller's parameters upon its performance. In their analysis, the authors show that the propagation of the airfoil drag coefficient to the propeller integral performance is very small.

Reuben et-al³ refer to the importance of airfoil at propellers, mainly at low Reynolds number. Their finds shed light on the 2-D conditions which these airfoils are operated. This emphasize the importance of high lift-to-drag properties, rather high lift capabilities.

Traub⁴ expanded the well-known Larabee's⁵ formulation of optimal designed propeller method. He introduced Reynolds-number dependency of the air-sections' 2-D aerodynamic characteristics. The results showed that

achieving a peak local radial lift-to-drag ratio along the blade does not appear to have a significant impact on the net efficiency of the propeller

Thus, the 2-D improvement of lift-to-drag ratio is less important than the induced characteristics of the propeller.

The importance of airfoil was demonstrated also for hovering-propeller, i.e. rotor. Vu et-al⁶ conducted numeric optimization of rotor blade at hovering flight condition. They started with standard NACA-0012 and constant chord blade. It is quite interesting that the optimized blade airfoil modifications are very small, while the chord distribution was radically altered to high tapered ratio blade.

There are places where airfoils are important such as in the case of Bingham⁷ paper. He evaluated airfoils for helicopter rotor, focusing mainly on their drag-divergence performance. Hence, mainly at off-design conditions of the blades' tip. At design conditions, the airfoils impact is quite small.

In a case of different "rotating wing" usage, Kanya and Visser⁸ discussed the importance of airfoil for horizontal axis wind turbine. In their research they found that the penalty for using common airfoil such as 4-digit NACA family over specialized airfoil is quite small.

In that sense, the current effort will concentrate on propeller performance influenced by airfoil drag (or drag-to-lift ratio). This is, as will be shown later, the major aerodynamic influence of the blade's air-section over the propeller performance.

Note that in addition to drag, there are many more considerations to be taken into account for airfoil selection. These can be summarized in the following list:

- a. Low pitching-moment. This is extremely important in helicopter rotors which high pitching moment can cause extreme fatigue stresses on the pitch-links and servos. This is the reason why rotor airfoils do not exhibit high camber, e.g. Ref. 9 which presents the UH-60 airfoil. Reference 10 takes this airfoil and tries to improve it using optimization which maintains its pitching moment characteristics with decreased drag. The results emphasize the small benefit of decreased drag with pitching moment penalty.
- b. High thickness due to structural considerations. This is mostly important for wind turbines blades¹¹ but also for root cross-sections in propellers and rotors. Structural properties of rotating wings cannot be compromised; thus thickness constraints are rigid as part of the design process.
- c. Trailing edge thickness due to manufacturing constraints. For composite/natural-composite propellers, the trailing edge thickness, especially close to the blade tip might constraint significantly the airfoil choice. This is more pronounce as the propeller diameter and chords decrease.
- d. High critical and drag-divergence Mach number characteristics. This requirement is important for high speed propellers or high rotational-speed/large radius blades.⁷ For hovering propeller or small sized propeller this is somewhat less important. Still, there are some exotic cases such as the Mars helicopter which this requirement is crucial.¹²
- e. Stall characteristics and maximum lift-coefficient is not a major requirement for propeller's air-sections. Although high lift-coefficient might be required at off-design flight regimes. In most cases, the design conditions try to exhibit moderate lift coefficient, in which the lift-to-drag ratio is optimal. Moreover, in propellers, the stall characteristics are very different from the 2-D characteristics due to the blade rotation. In most cases, stall is delayed to much higher angles of attack.¹³

Saying that, the current effort will emphasize the propeller performance and its dependency on the cross sectional drag. This will lead to the reason for the relatively lack of importance considering of airfoil choice for propellers. Some numeric examples will demonstrate the impact of “good” versus “excellent” airfoil choice and its influence on the overall propeller performance.

III. Propeller Thrust and Power

To estimate the importance of propeller’s air-sections one should refer to the thrust and required power. Propeller is used to produce thrust, T . The penalty for this thrust is the required power, P , as invested in the propeller axis. Propeller performance can be characterized as the amount of P for required T or the amount of produced T for available P .

In what follows, only axial flight is considered, hence flight velocity, V , coincide with the propeller rotation axis. For this flight conditions, required power is consisted of three contributions:

- a. Propulsive thrust power
- b. Induced power
- c. Cross-sectional drag power

Propulsive thrust power is the product of thrust to flight velocity, $T \cdot V$. The induced power is dependent on the induced velocity, v_a , which the propeller produces while accelerating the air. This gives an induced power which is the product of thrust with induced velocity, $T \cdot v_a$. Cross-sectional drag power, P_{Cd} , represents the power required to overcome the parasite 2-D drag of the blade’s cross sections. This drag is represented through the cross-sectional drag coefficient, C_d .

Total required power, P , can be written as the sum of the above contributions as depicted in Eq. (1).

$$P = T \cdot (V + v_a) + P_{Cd} \quad (1)$$

The propeller efficiency, η , is defined as the ratio between the propulsive thrust power to the total required power:

$$\eta = \frac{T \cdot V}{P} = \frac{1}{1 + \frac{v_a}{V} + \frac{P_{Cd}}{T \cdot V}} \quad (2)$$

Note that for static operation, $V=0$, the efficiency is equal zero. For hover-propeller a substitute parameter is defined, Figure-of-Merit, FM . It is defined as the ratio between the ideal induced-power to total power, which is somewhat different from efficiency definition:

$$FM = \frac{T \cdot v_{a,ideal}}{P} = \frac{T \cdot v_{a,ideal}}{T \cdot v_a + P_{Cd}} \quad (3)$$

These two metrics, η and FM , will be used separately for axial-flight and hovering propeller, respectively. These terms are to be simplified, thus enabling estimation of the airfoil aerodynamic characteristics impact.

The ideal axial induced velocity, $v_{a,ideal}$, is given using simple momentum theory as developed by Glauert:¹⁴

$$v_{a,ideal} = -\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2 \cdot \rho \cdot A}} \quad (4)$$

Note that $v_{a,ideal}$ represents the minimal induced velocity. This was presented in various occasions^{18,19,20}, hence Eq. (4) proved to under-predicts the actual axial-induced velocity. This fact can be treated by using a correction factor $\kappa_a > 1$, which relates between the actual v_a , and $v_{a,ideal}$.

$$v_a = \kappa_a \cdot v_{a,ideal} \quad (5)$$

As shown in Fig. 1, the actual induced velocity is higher compared to the ideal. In this figure, the axial flight velocity, V , and axial induced velocity, v_a , are normalized by the hover induced velocity, $v_{a,h}$.

At normal axial operation of propellers, where V is high, thus $V/v_{a,h} \geq 1.5$, κ_a is relatively close to unity. As speed decreases, the induced velocity deviates from the axial momentum ideal estimation. At hover the correction factor reaches to $\kappa_a = 1.2 \div 1.3$.

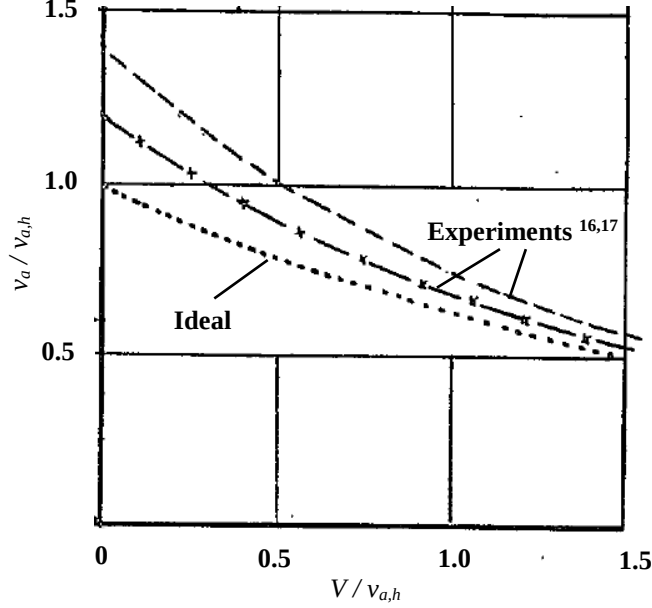


Fig. 1, Ideal induced velocity versus experimental results. Taken from Ref. 15

Using Eq. (4) for static condition, $V=0$, and substituting into FM definition in Eq. (3) gives the well-known relation between propeller thrust and required power at hover:

$$FM = \frac{T^{1.5}}{P \cdot \sqrt{2 \cdot \rho \cdot A}} \quad (6)$$

Cross-sectional drag power can be estimated using an equivalent drag coefficient, C_{d0} , which represents the average 2-D parasite drag of the airfoils. Using blade-element model, BEM,²⁰ the cross-sectional drag power, P_{Cd} , is:

$$\frac{P_{Cd}}{\rho \cdot A \cdot V_{tip}^3} = \frac{\sigma \cdot C_{d0}}{8} \quad (7)$$

where σ is the propeller solidity, i.e. the ratio between the blade area to disk area:

$$\sigma = \frac{N_b \cdot c_{av} \cdot R}{A} \quad (8)$$

N_b is number-of-blades and c_{av} is the average blade chord.

Note that the equivalent drag coefficient, C_{d0} , is not the average of the 2-D drag-coefficient distribution, C_d , but the average of weighted $r^3 \cdot C_d$:

$$C_{d0} = \frac{4}{R^4} \cdot \int_0^R C_d(r) \cdot r^3 \cdot dr \quad (9)$$

In propellers, similar measure for the blade-solidity, σ , is the Activity-Factor, AF , which is defined, following Eq. (9), with chord weighted by r^3 . Note that the factor 100,000/32, in this expression, is required to make AF 's value at the ball-park of 100.

$$AF = \frac{100,000}{32 \cdot R^4} \int_0^R \frac{c}{R}(r) \cdot r^3 \cdot dr \quad (10)$$

In what follows, propeller disk area, A , air density, ρ , and blade tip velocity, V_{tip} , are used as normalization factors. V_{tip} is defined as the tangential velocity of the blade tip at static condition, $V=0$:

$$V_{tip} = \Omega \cdot R \quad (11)$$

where Ω is the propeller angular velocity and R is the propeller radius.

The thrust and power coefficient, C_T and C_P , respectively, are defined using these normalization factors:

$$C_T = \frac{T}{\rho \cdot A \cdot V_{tip}^2} \quad C_P = \frac{P}{\rho \cdot A \cdot V_{tip}^3} \quad (12)$$

In propeller, thrust produced through the cross-sectional lift. Using again BEM and assuming a representative design lift coefficient, C_{l0} :

$$C_T = \frac{\sigma \cdot C_{l0}}{6} \quad (13)$$

Here, similar to C_{d0} , the design lift coefficient, C_{l0} , is defined using r^2 weighting:

$$C_{l0} = \frac{3}{R^3} \cdot \int_0^R C_l(r) \cdot r^2 \cdot dr \quad (14)$$

All of the above relations are now used to simplify the efficiency, η , and figure-of-merit, FM , expressions.

IV. η influenced by 2-D aerodynamics

Substituting Eqs. (7) through (13) to Eq. (2) gives an estimation for the propeller efficiency.

$$\eta = \frac{1}{\frac{1}{2}(2 - \kappa_a) + \kappa_a \cdot \sqrt{\frac{1}{4} + \frac{\pi^3}{3} \cdot \frac{\sigma \cdot C_{l0}}{J^2}} + \frac{3\pi}{2} \frac{1}{J \cdot C_{l0}/C_{d0}}} \quad (15)$$

where J is the advance ratio defines as:

$$J = 2\pi \frac{V}{V_{tip}} \quad (16)$$

For propellers, as flight speed increases, the induced factor, κ_a , becomes unity. This gives a good approximation for propeller efficiency at design conditions.

$$\eta = \frac{1}{\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{\pi^3}{3} \cdot \frac{\sigma \cdot C_{l0}}{J^2}} + \frac{3\pi}{2} \frac{1}{J \cdot C_{l0}/C_{d0}}} \quad (17)$$

As J increases, η reaches unity. This is true due to the increased part of the propulsive thrust power, thus the induced and cross-sectional drag power become negligible.

The optimum efficiency is considered for minimum lift-to-drag, as shown from the term of C_{l0}/C_{d0} . One should remember that C_{l0} is not arbitrary. The required thrust dictates the design lift-coefficient through the relation in Eq. (13). This makes the design of a propeller a combination of the correct choice of its solidity, σ , which brings the 2-D lift coefficient, C_{l0} , to the proper lift-to-drag region.

Equation (17) suggests that for given geometry, and thrust demand, the change of airfoil is expressed only in the lift-to-drag ratio.

Fig. 2 shows the 2-D aerodynamic characteristics for NACA-0012 airfoil. The pick lift-to-drag is at about $C_l = 0.8$ and it reaches $C_l/C_d = 75$.

Fig. 3 demonstrates the influence of lift-to-drag ratio on the propeller efficiency. Even increasing the lift-to-drag ratio to 150, efficiency increases by about 1% compared to the efficiency at $C_{l0}/C_{d0} = 75$. In addition, compared to the lift-to-drag ratio, advance ratio has much substantial influence.

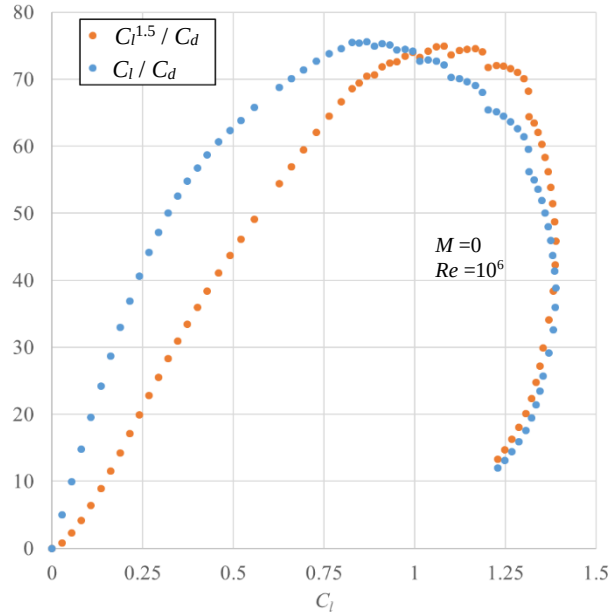


Fig. 2, NACA-0012 aerodynamic characteristics, source: <http://airfoiltools.com/>

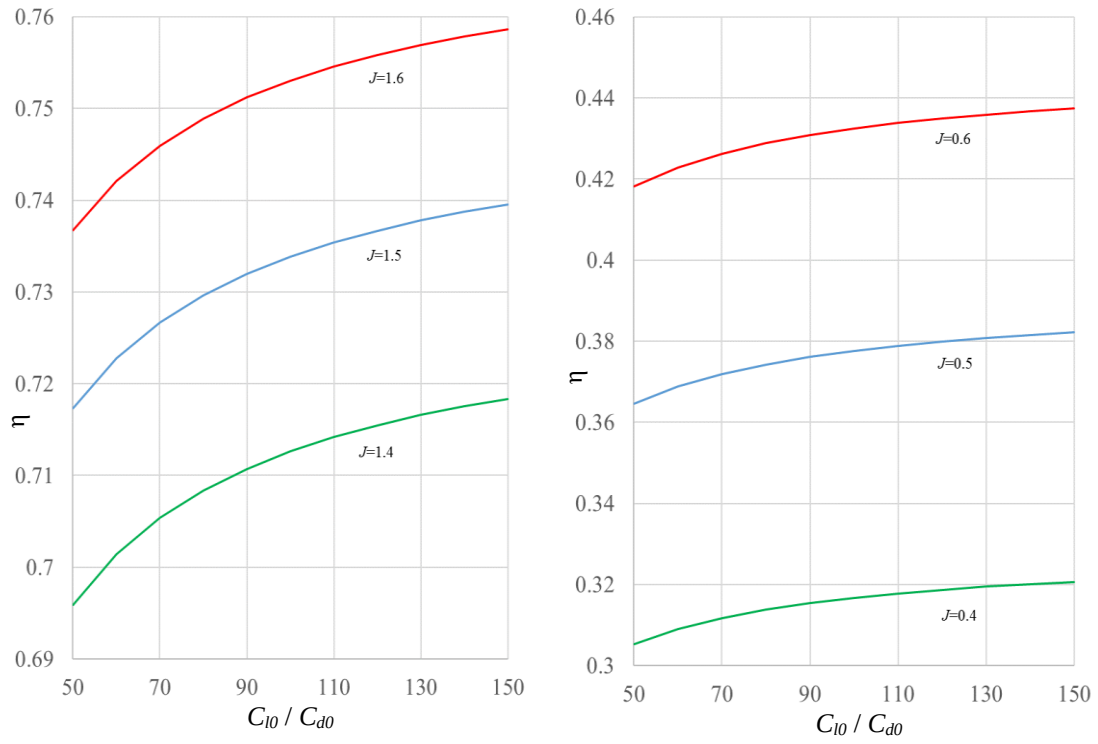


Fig. 3, Propeller efficiency estimations, $\sigma=0.04$
Note the zoomed vertical scale which allows noticing C_{l0} / C_{d0} influence

V. FM influenced by 2-D aerodynamics

Substituting Eqs. (11) through (13) to Eq. (3) gives an estimation for the propeller Figure-of-Merit at static conditions, $V=0$:

$$FM = \frac{1}{\kappa_a + \frac{3\sqrt{3}}{2} \frac{1}{\sqrt{\sigma}} \frac{1}{C_{l0}^{1.5} / C_{d0}}} \quad (18)$$

This result is somewhat more elegant compared to the efficiency result of Eq. (15) due to static condition constraint. Moreover, the 2-D aerodynamic parameter of $C_{l0}^{1.5} / C_{d0}$ is noticeable. As shown in Fig. 2, the aerodynamic 2-D parameter reaches $C_{l0}^{1.5} / C_d = 75$ for the NACA-0012. For perfect blade, without drag, the FM get close to unity, as dictates by the induced factor, κ_a .

Left chart of Fig. 4 shows the influence of $C_{l0}^{1.5} / C_{d0}$ on FM . Again, substantial increase or decrease with $C_{l0}^{1.5} / C_{d0}$ influences in moderate manner. For example, an increase from 70 to 100 (about 50% increase of $C_{l0}^{1.5} / C_{d0}$) shall result in a change to the FM of about 2%. As FM become lower, influence of $C_{l0}^{1.5} / C_{d0}$ increases, but still it is moderate.

Better insight can be substantiated from the right chart which shows the ratio between the cross-sectional drag power, P_{Cd} , to the total required shaft power, P . One can find the relation between the two charts through the following equation:

$$\frac{P_{Cd}}{P} = 1 - FM \cdot \kappa_a \quad (19)$$

According to the right chart of Fig. 4, influence of the airfoil aerodynamic characteristics is low. It suggests that a ratio of 1:5 up to 1:20 is exists between airfoil-drag and total required power. This means that improvement with the 2-D airfoil aerodynamics is translated to much lower influence on the propeller performance - this emphasizes the low impact of the sectional drag.

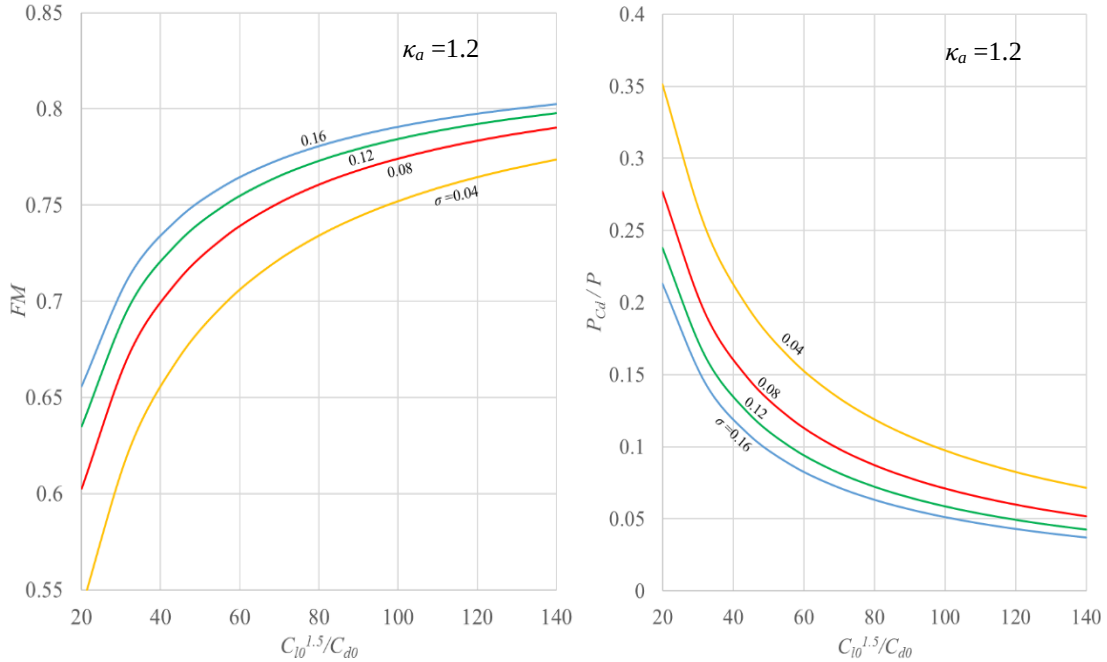


Fig. 4, Hover propeller performance estimations

VI. Hovering-propeller demonstration

To demonstrate the above results, a well analyzed propeller is used – Mejzlik 18×6.²¹ Note that this demonstration aims merely to strengthen the above analytic relations. One can use these relations in a general manner, demonstrating the 2-D influence on other propeller geometries and flight regimes.

Fig. 5 shows the Mejzlik 18×6 propeller and Fig. 6 shows its geometric properties as function of radial coordinate, r , i.e. pitch, β , chord-to-radius ratio, c/R , and thickness-ratio, t/c , distribution. The propeller radius is $R = 0.23$ m. It has 2 blades and a solidity of $\sigma = 0.066$.

At design conditions, the Mejzlik 18×6 produces thrust of $T = 2.8$ kgf which is established as the required thrust for hover (static operation). All analyses are done for sea-level, ISA.

The same blade geometry is replicated for 2, 3, and 4 bladed propellers, thus range of solidities, σ , are to be explored. Table 1 contains the performance of these propellers, as resulted from BEM analysis. From the resulted thrust and power, the figure-of-merit, FM , and P_{Cd}/P are calculated using Eq. (6).

Fig. 7 shows the distribution of Reynolds and Mach number. Reynolds number is $Re \sim 100,000 \div 150,000$. This is quite low, thus the airfoils exhibit relatively high drag, especially at the blade-tip, where the thickness-ratio is high and Reynolds number is low - this might be one of the reasons for the relatively low FM . Note that the Mejzlik 18×6 propeller is a typical small propeller and consider quite adequate for hovering purposes. Hence, the results here reflect the situation in most cases of small propellers. For large propellers, where the 2-D characteristics are better, the improvement potential is even lower.

Table 1, Mejzlik 18×6 performance for $T=2.8$ kgf

	σ	Ω , rpm	P , W	FM	$P_{Cd}/P^{(*)}$	C_{l0}	$C_{l0}^{1.5}/C_{d0}$
2 Blades	0.0663	5,010	333	0.679	0.117	0.850	58.5
3 Blades	0.0994	4,490	348	0.650	0.155	0.704	34.6
4 Blades	0.132	4,220	362	0.624	0.188	0.600	23.7

(*) $\kappa_a = 1.3$ is assumed

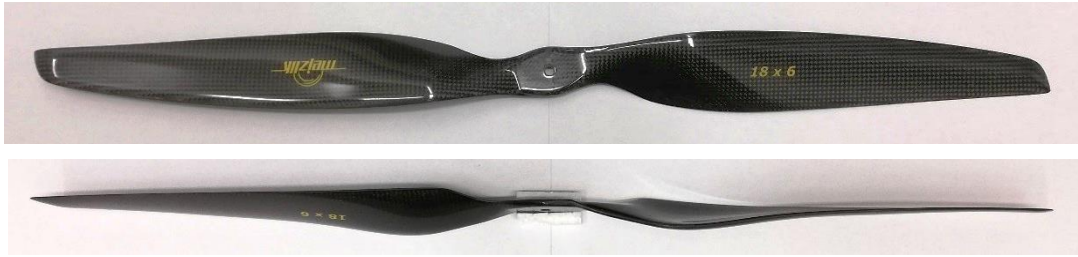


Fig. 5 Mejzlik 18×6 Propeller, front and top views

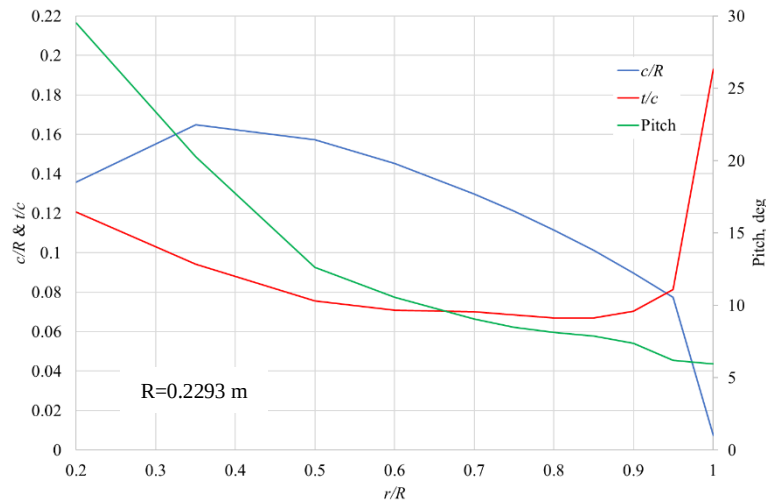


Fig. 6 Mejzlik 18×6 Propeller, geometric properties

The 2-D properties of C_{l0} , and $C_{l0}^{1.5} / C_{d0}$, are depicted in Table 1. Actual sectional distribution of these properties are shown in Fig. 8. Notice that Table 1 values seem somewhat conservative compared to the actual distribution in Fig. 8. It might be due to accumulating influence of two reasons:

- The actual induced velocity is higher by about 30%, hence $\kappa_a \sim 1.3$. Right chart in Fig. 8. shows the actual distribution of κ_a .
- Tip sections exhibits higher drag-coefficient due to much lower Reynolds number and high thickness-ratio. The drag coefficient is weighted according to r^3 - see Eq. (9), thus the equivalent drag coefficient is higher.

Still, the estimated calculation of the radius averaged C_{l0} , and $C_{l0}^{1.5} / C_{d0}$, as presented in Table 1, fits fairly well to the actual distributions as shown in Fig. 8.

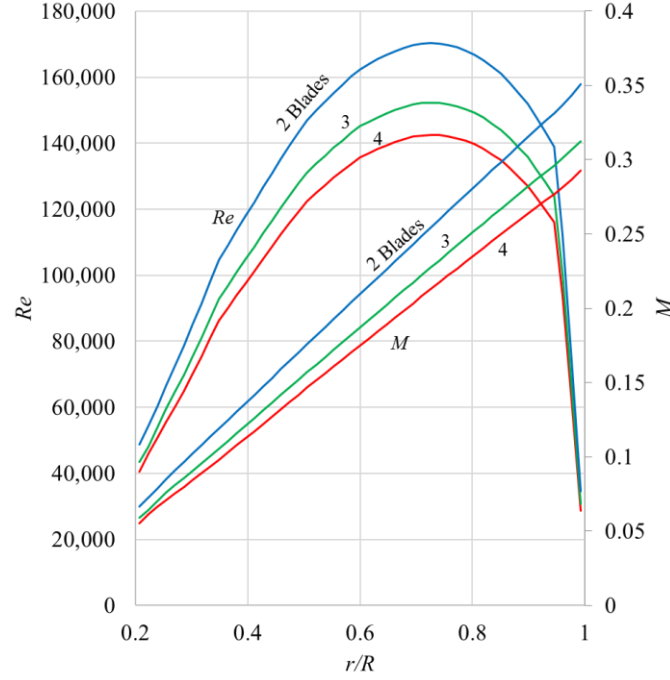


Fig. 7, Mejzlik 18×6 Reynolds and Mach number distribution, T=2.8 kgf

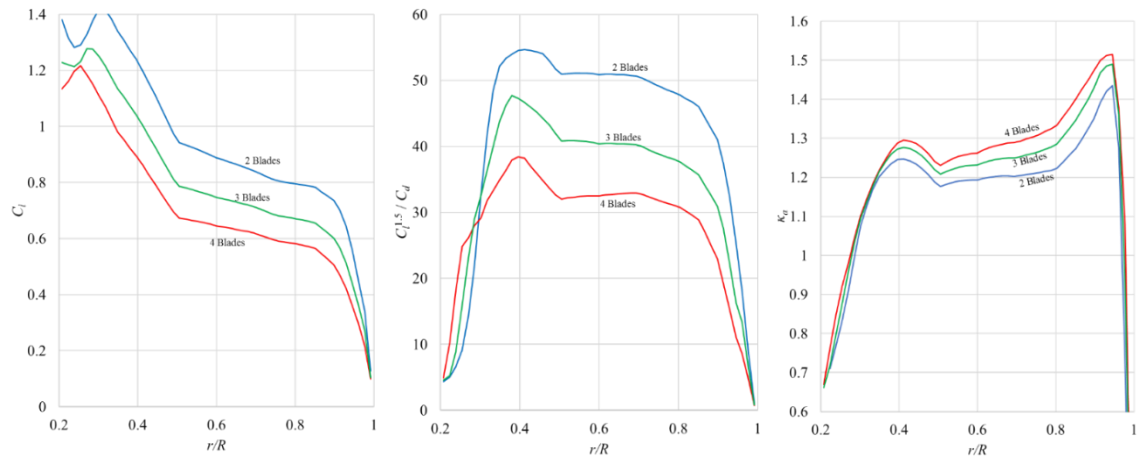


Fig. 8, Mejzlik 18×6 sectional aerodynamic properties distribution

To examine potential impact of an improved airfoil, increased and decreased $\Delta C_l^{1.5}/C_d$ are implemented. The same analysis was repeated with factor implemented over drag coefficient. The results are presented in Table 2.

Although the airfoil aerodynamic, namely $\Delta C_l^{1.5}/C_d$, is changed by $\pm 25\%$, the resulted power is influenced by merely $3\% \div 5\%$ while FM is changed by about 3% . This is somewhat disappointing considering the large changes with the airfoil aerodynamics.

This demonstrates the limited influence of airfoil aerodynamic characteristics over propeller performance. Note, that for the current case with relatively low Reynold number, airfoil improvement potential is high. For other cases, which are already starting with high FM , the influence is expected to be even less pronounce.

Table 2, Mejlzik 18x6 performance influence by $C_l^{1.5}/C_d$, for $T=2.8$ kgf

N_b	$\Delta C_l^{1.5}/C_d$	Ω , rpm	P , W	ΔP , %	FM
2 Blade	-25%	5,020	350	5.2	0.645
	0	5,010	333	0	0.679
	+25%	5,000	323	-3.1	0.701
3 Blades	-25%	4,500	366	5.2	0.618
	0	4,490	348	0	0.650
	+25%	4,490	337	-3.1	0.671
4 Blades	-25%	4,230	382	5.4	0.592
	0	4,220	362	0	0.624
	+25%	4,210	350	-3.2	0.645

VII. Summary

This paper focuses on propeller's air-sectional 2-D aerodynamic properties. The air-section selection is based on numerous criteria. It is shown that considering the propeller performance, most crucial criterion is the lift-to drag ratio, or a similar metric based mainly of the cross-sectional drag. Moreover, this metric influences the performance in limited way - its impact is small.

The reason for this low influence is the source of propeller required power. Airfoil drag is one of 3 sources at axial flight and 2 sources at hover. During axial flight most of the required power is due to the propulsive power and during hover most of the required power is due to the induced power. The airfoil drag has low influence on the total required power.

The paper explores a general propeller form, with no dependency on size nor flight conditions. For this general propeller, at axial flight, changing the drag by 25% influences its efficiency by less than 1%.

Similar dependency is shown for hover performance. Changing the drag by 25% influences figure-of-merit by 3% or less. This demonstrates the low impact of airfoil-drag characteristics.

To demonstrate the correctness of these general results, Mejlzik 18x6 is used. It is a well-known, validated, of-the-shelf propeller. By pronounced change of its 2-D properties, a negligible change of its performance is demonstrated. Although this is merely an example, one can use the general analytic terms and validate the paper conclusions for other cases, involving different propellers in various flight conditions.

One should remember that the drag or lift-to-drag ratio of the airfoil is only a single metric out of many others. For example, thickness distribution should be considered mainly due to structural constraints. For propellers, it seems that the 2-D drag properties are not very important as-long-as the airfoil has reasonable drag characteristics. Hence, using optimization to design propeller to best performance by geometric shaping of its air-sections is not relevant. One should choose reasonable existing air-section and focus on the pitch/chord distributions.

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